Q2

The input, target, and weight matrices are:

$$X = [1, 2], \qquad T = [0, 0]$$

$$W_1 = \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix}, W_2 = \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix}$$

Forward Propagation:

$$S_2 = XW_1 = [1, 2] \begin{bmatrix} 2 & 1 \\ 0 & 1 \end{bmatrix} = [2, 3]$$

$$Y_2 = \mathrm{ReLU}(S_2) = [2, 3]$$

$$S_3 = Y_2 W_2 = [2, 3] \begin{bmatrix} 1 & 2 \\ 1 & 2 \end{bmatrix} = [5, 10]$$

$$Y_3 = \mathrm{ReLU}(S_3) = [5, 10]$$

The MSE loss is:

$$L = \frac{(5-0)^2 + (10-0)^2}{2} = 62.5$$

Gradient Calculation for W_2:

$$\frac{\partial L}{\partial Y_3} = Y_3 - T = [5 \quad 10]$$

Since both elements of (S_3=[5,10]) are positive, the ReLU derivative is (1) for both elements. Therefore,

$$\frac{\partial L}{\partial S_3} = [5 \quad 10]$$

$$\frac{\partial L}{\partial W_2} = Y_2^T \frac{\partial L}{\partial S_3}$$

$$\frac{\partial L}{\partial W_2} = \begin{bmatrix} 2 \\ 3 \end{bmatrix} [5 \quad 10] = \boxed{\begin{bmatrix} 10 & 20 \\ 15 & 30 \end{bmatrix}}$$

Gradient Calculation for W_1:

$$\frac{\partial L}{\partial Y_2} = \frac{\partial L}{\partial S_3} W_2^T$$

$$\frac{\partial L}{\partial Y_2} = [5 \quad 10] \begin{bmatrix} 1 & 1 \\ 2 & 2 \end{bmatrix} = [25 \quad 25]$$

Since both elements of (S_2=[2,3]) are positive, the ReLU derivative is (1) for both elements. Therefore,

$$\frac{\partial L}{\partial S_2} = [25 \quad 25]$$

$$\frac{\partial L}{\partial W_1} = X^T \frac{\partial L}{\partial S_2}$$

$$\frac{\partial L}{\partial W_1} = \begin{bmatrix} 1 \\ 2 \end{bmatrix} [25 \quad 25] = \boxed{\begin{bmatrix} 25 & 25 \\ 50 & 50 \end{bmatrix}}$$